

Predictors of the macroinvertebrate fauna in semiarid aquatic systems

Tentativa de manuscrito para a NENT*

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1 Introduction

2 Material and Methods

2.1 Study Area and Sampling Design

The present study was performed at the eastern limits of the semiarid region of Brazil (Figure 1). This area is characterized by low average precipitation, concentrated in a few months of the year, usually between January and July (Figure 2), and high average annual temperatures between 20 and 32 °C (Barbosa et al. 2025). The main hydrological feature in the study area is the intermittence of surface water flow of its streams and rivers and, as a consequence, many man-made reservoirs built from the damming of the intermittent streams (Maltchik and Medeiros 2006). These rivers flow through the “Caatinga”, a deciduous arboreal to shrubby open forest (Silva et al. 2017). The climate in the study area is semiarid (BS’h hot and dry) and equatorial (As, dry summer) (Peel et al. 2007). A diverse array of sampling sites was chosen to represent the most common aquatic environment types of semiarid Brazil, with different sets of specificities and across different catchment basins. We sampled six sites, three of which consisted of stream reaches with surface water flow (during the rainy season) or isolated temporary pools (during the dry season) and three sites in artificial reservoirs created from stream impoundment. Sampling was conducted during the year of 2006 on four occasions during the rainy (April and June) and dry seasons (September and December).

2.2 Data Collection

Environmental characteristics of each site were measured in four sets of variables: (a) site morphology, (b) water quality, (c) sediment composition, and (d) marginal habitat structure. Site morphology was assessed by their width (cm) and depth (cm) measured from three random transects. Catchment scale variables (such as elevation and river length) were measured using handheld GPS and satellite imagery. Water quality was evaluated as physical and chemical variables that were measured using portable equipment for temperature ($^{\circ}\text{C}$) and dissolved oxygen (mg/L). Transparency (cm) was measured using a Secchi disk and water velocity (m/s) was estimated using the float method (Maitland 1990). Sediment composition and the habitat physical structure followed protocols from Pusey Pusey et al. (2004) and Mugodo et al. (2006), adapted by Medeiros et al. (2008), where they are estimated from 9 to 12 one meter quadrants along the margins (terrestrial-aquatic interface) and determined by visual estimation of sediment type percentage cover (mud, sand, cobbles, small gravel, large gravel, rocks and bedrock) and of littoral and sub-aquatic structures (macrophytes, littoral grass, leaf litter, attached algae, filamentous algae, overhanging vegetation, submerged vegetation, small woody debris, large woody debris and root masses). At each study site, three samples of benthic macroinvertebrates were collected in the margins using a D type net (40 cm wide and 250 m) to represent different habitat types. Samples were fixed in situ with 4% formalin and subsequently preserved in 70% ethanol. Macroinvertebrate specimens were sorted and identified to the lowest possible taxonomic level, usually family (McCafferty and Provonsha (1983), Mugnai et al. (2010), Thorp and Covich (2010) and Merritt et al. (2019), among others).

2.3 Statistical Analyses

Prior to statistical analysis, environmental variables were checked for multivariate collinearity and square root transformed to enhance normality and homogeneity of variances (Sokal and Rohlf 1995).

Fig.: Environment variables pairs

Fig.: Habitat variables correlations plot

These variables were then subject to a Principal Component Analysis (PCA) to have their multivariate correlations described across sites. The PCA was conducted on the square root transformed and scaled data using the R package FactoMineR (Lê et al. 2008).

For the macroinvertebrate community, all analyses were performed on density of individuals (ind/m^2) calculated by the ratio between the number of individuals and the sampled area of the D-net in each sample. Macroinvertebrate fauna was described by means of average density and rarefied taxa richness, where richness was standardized by the average sample size across all samples. Life stages (larvae, pupae, nymph and adults) were treated as separate taxa, considering their ecological differences (Ross et al. 1987). Comparisons of density and

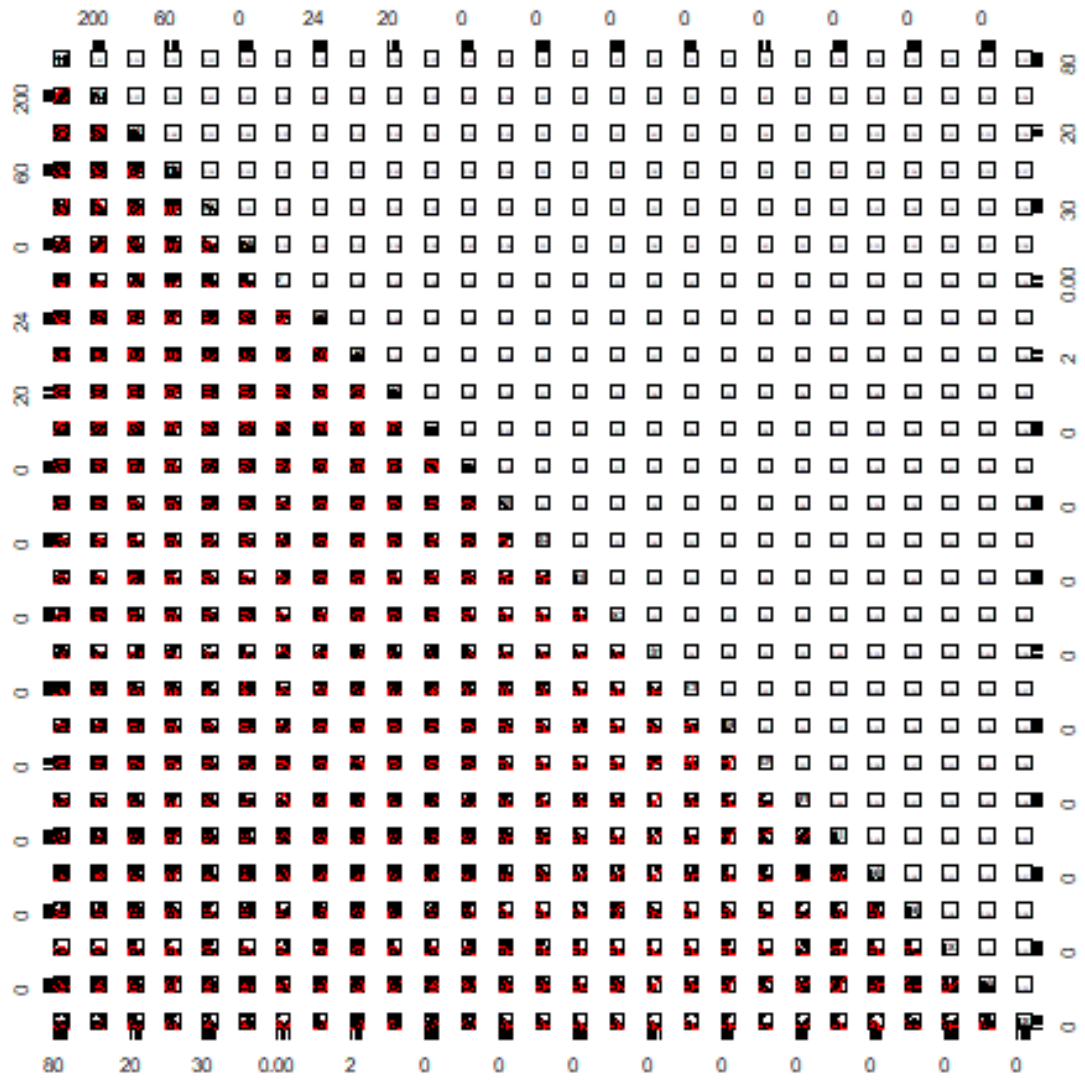


Figure 1: Environment variables correlations pairs.

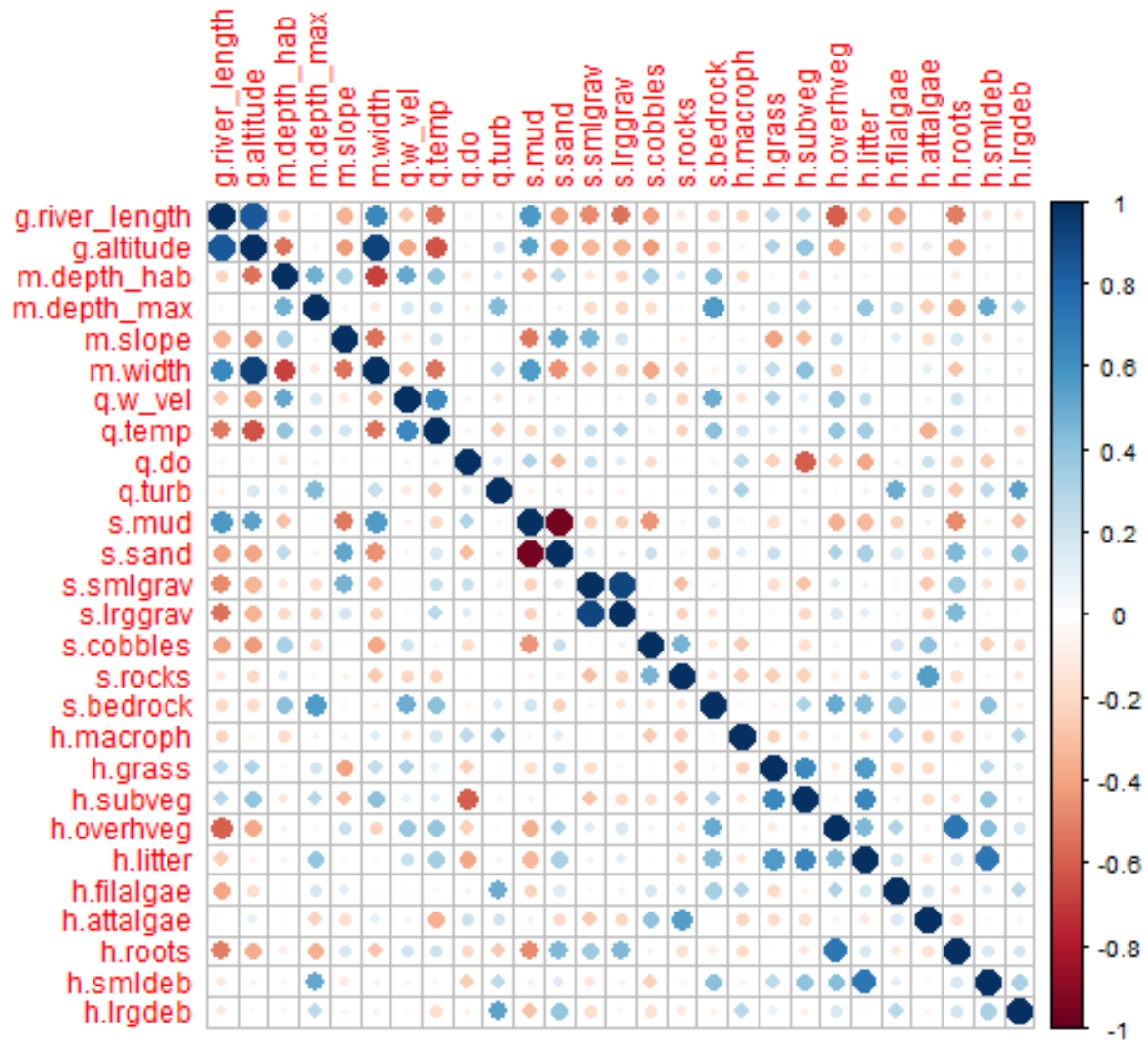


Figure 2: Habitat variables correlations correlogram.

richness among study sites were tested by one-way ANOVA followed by post hoc multiple comparisons using Tukey's HSD test ($\alpha=0.05$) (Zar 1999).

A Stepwise Multiple Regression (SMR) analysis using the Akaike Information (MASS package, stepAIC function, Venables and Ripley (2002)) as the selection criterion was performed in both directions (forward and backward) to identify the best predictors across subsets of variables (site morphology, water quality, sediment composition and marginal habitat structures) for species richness and density (Vendramin et al. 2020).

Changes in species composition among sites were ordinated using Non-metric Multidimensional Scaling (NMDS) (Lanés et al. 2018). Data were arcsine square-root transformed after relativization and the Bray-Curtis distance was calculated.

Significance of group formed by ordination was tested by Multi-Response Permutation Procedure (MRPP) (BIONDINI et al. 1985, McCUNE and GRACE 2002). For all MRPP analyses, an A value is presented as a measure of homogeneity degree in a cluster, as compared to random expectation. When significant differences were shown by MRPP, Indicator Species Analysis was performed with the aim of revealing which macroinvertebrate taxa contributed for cluster separation. The index value (IV) is calculated according to the method proposed by Dufrene and Legendre (DUFRENE and LEGENDRE 1997). This value was then tested for statistical significance by means of Monte Carlo method (1000 repetitions).

We performed a CCA/dbRDA for each site to see which environmental/habitat variables were the most important for determining each benthic community composition.

We used partial distance-based redundancy analysis (dbRDA; Legendre and Anderson (1999)) with the vegan package (version 2.5-7; Oksanen et al. (2020)) in R to test the effects of the hydration events on the composition of invertebrates that emerged from the sediment. All statistical analyses were performed in the statistical environment of R v.2.9.0 (Team 2017).

For determining the relation between fauna composition and environmental variables, a Canonic Correspondence Analysis (CCA) was performed (McCUNE and GRACE 2002). Data matrix was centered and normalised, and correlations were tested by Monte Carlo method with 999 repetitions. Environmental variables used at CCA were as follows: flow, altitude, mud, sand, macrophytes, submerged vegetation, leaf litter, vegetal debris and dissolved oxygen (see FARIAS et al. 2012b, Carvalho et al. 2013b). Density matrices and environmental variables were transformed by Log10 (n+1) (Sokal and ROHLF 1969, Maltchik et al. 2010). Statistical analyses were performed with SPSS 13.0 (SHERIDAN and LYNDALL 2001) and PC-ORD 4.27 packages (McCUNE and MEFFORD 1999).

3 Results

3.1 Environmental variables

The study sites showed a wide array of environmental features, as expected from their different types (Figure 3). Water flow was present only in smaller streams in lower altitudes (sites 4-6) and during the rainy season (0.10-0.17 m/s). Stream sites tended to be narrower with widths ranging from 5.4 to 29.6 m, compared with 72.2 to 330 m for reservoirs. Littoral depths varied widely from 4.7 to 81.3 cm across all sites. Dissolved oxygen (DO) ranged from 1.8 to 9.4 mg/L and temperatures from 24.0 and 35.2 °C. Turbidity was low with Secchi depths reaching 90 cm, even though some sites had depths as low as 16 cm of the Secchi disk. Mud and sand were the main substrates across all sampled sites (with average covers of 54.4 and 37.9%, respectively). The habitat elements that gave greater overall contributions were attached and filamentous algae (12% on average), littoral grass (9.0%) and aquatic macrophytes (8.4%), but these contributions varied widely across sites and season (Figure 3) (detailed information on the study sites has been published elsewhere by Medeiros et al. 2008a, Medeiros et al. 2024).

Fig.: Environment data GT table

Principal Component Analysis described the overall structure of the study sites and the most important features in separating them in terms of their physical and chemical variables, site morphometry, sediment composition, and marginal habitat structure (Figure 4). PCA explained 41.9% of the variance in the environmental variables, with the first axis (27.6%) showing a gradient from large reservoirs to smaller streams sites. Large scale morphology variables such as altitude, river length and site width were important to describe most reservoirs, whereas river sites were better described by local physical, chemical and habitat variables, such as water velocity, dissolved oxygen, temperature and overhanging vegetation. Larger reservoirs were mostly associated with a muddy substrate whereas river sites included a more diverse array of substrates including sand, rocks and gravel and cobbles. Other variables such as submerged vegetation and leaf litter (habitat structure), and slope (morphology) were important in explaining specific sites at given sampling occasions.

Fig.: Environment data PCA - fviz

Fig.: Environment data PCA - R Base

3.2 Benthic macroinvertebrates

A total amount of 28155 individuals was collected, divided into 32 taxonomic groups. Out of these, Thiaridae (1340.15 ind./m² ± 3068.95), larvae of Chironomidae (629.07 ind./m² ± 1424.72), and Oligochaeta (380.49 ind./m² ± 856.62) were the most the representative (Table 2).

Fig.: Density data table

	Buique Acude MU	Buique Acude SA	Buique Rio EP	Serido Acude RE	Serido Rio CI	Serido Rio SE
g.river.length	214 (214-214)	212.7 (212.7-212.7)	196.8 (196.8-196.8)	110.2 (110.2-110.2)	83 (83-83)	163.2 (163.2-163.2)
g.altitude	725 (725-725)	713 (713-713)	402 (402-402)	270 (270-270)	169 (169-169)	226 (226-226)
m.depth.hab	29.7 (22-41.3)	6.7 (4.7-8.2)	50.7 (49.3-52.8)	36.2 (22.3-54.7)	33.3 (22.7-45.3)	53.2 (32.3-81.3)
m.depth.max	114.7 (112-117)	65 (60-69)	95 (80-110)	117 (87-154)	67.8 (60-79)	98.8 (74-110)
m.slope	30 (30-30)	30 (30-30)	90 (90-90)	60 (60-60)	60 (60-60)	30 (30-30)
m.width	240.4 (234.5-247.6)	313.7 (289.5-330)	25.6 (20-29.6)	90.6 (72.2-102)	15.4 (10.7-18.5)	11.8 (5.4-19.6)
q.w.vel	0 (0-0)	0 (0-0)	0 (0-0)	0.025 (0-0.1)	0.04 (0-0.159)	0.073 (0-0.167)
q.temp	27 (26-29)	26.7 (24-29.2)	29 (28.9-29)	31.6 (29-34)	30.4 (27.6-35.2)	31.4 (28.3-32.9)
q.do	4.9 (1.8-7.3)	6.1 (1.9-8.8)	5.2 (5-5.6)	7.1 (4.8-9.4)	4.9 (3-6.9)	6 (5.4-6.5)
q.turb	65 (43-89)	42.1 (25.7-55)	37.4 (30-50)	67.4 (51.7-90)	42.3 (26-60)	30.8 (16-46)
s.mud	59.1 (20.6-91.8)	94.4 (87.8-98)	40.5 (33.7-48.9)	48.1 (5-81.7)	17.8 (0.7-48.8)	66.4 (40-95)
s.sand	38.4 (3.2-77)	3.9 (2-6.7)	55.6 (47.9-63)	45.5 (8.3-95)	60 (22.5-87.7)	23.7 (1.8-40)
h.macroph	0 (0-0)	12.8 (0-46.7)	0 (0-0)	35.7 (5.8-54.8)	0 (0-0)	2.1 (0-8.3)
h.grass	29.3 (5.6-54)	5.2 (0-13.3)	0.7 (0-2)	2.1 (0-8.1)	5.8 (0-23.3)	11.1 (0.1-20)
h.subveg	15 (0-36.6)	7.3 (0-26)	0 (0-0)	4.2 (0-16.7)	0.8 (0-3)	2.5 (0-10)
h.overhveg	0 (0-0)	0.9 (0-3.3)	0 (0-0)	10.4 (0-33.3)	17.1 (0-33.3)	0 (0-0)
h.litter	2.1 (0.4-5)	0 (0-0)	0.4 (0.3-0.5)	1.5 (0.6-3.7)	1.5 (1-2.3)	0.4 (0-0.8)
h.roots	0 (0-0)	0 (0-0)	0 (0-0)	0 (0-0)	2.3 (0-5)	0 (0-0)
s.gravel	0.8 (0-2.4)	1.7 (0-5.6)	3.1 (2-4)	4.3 (0-13.3)	19.7 (11.7-28.8)	6.3 (0.3-20)
s.rock	1.7 (0-5)	0 (0-0)	0.8 (0-1.2)	2.1 (0-8.3)	2.5 (0-10)	3.6 (0-8.3)
h.algae	10.1 (0-30)	17.3 (0-43.3)	0.1 (0-0.4)	17.7 (0-33.3)	21.4 (0-75)	5.1 (0-12)
h.debris	10.7 (10-11.7)	0.9 (0-3.8)	1.1 (0.4-2.5)	7.2 (1.7-13.7)	4.7 (1-10.3)	2.6 (0-5)

Figure 3: Environment data GT table

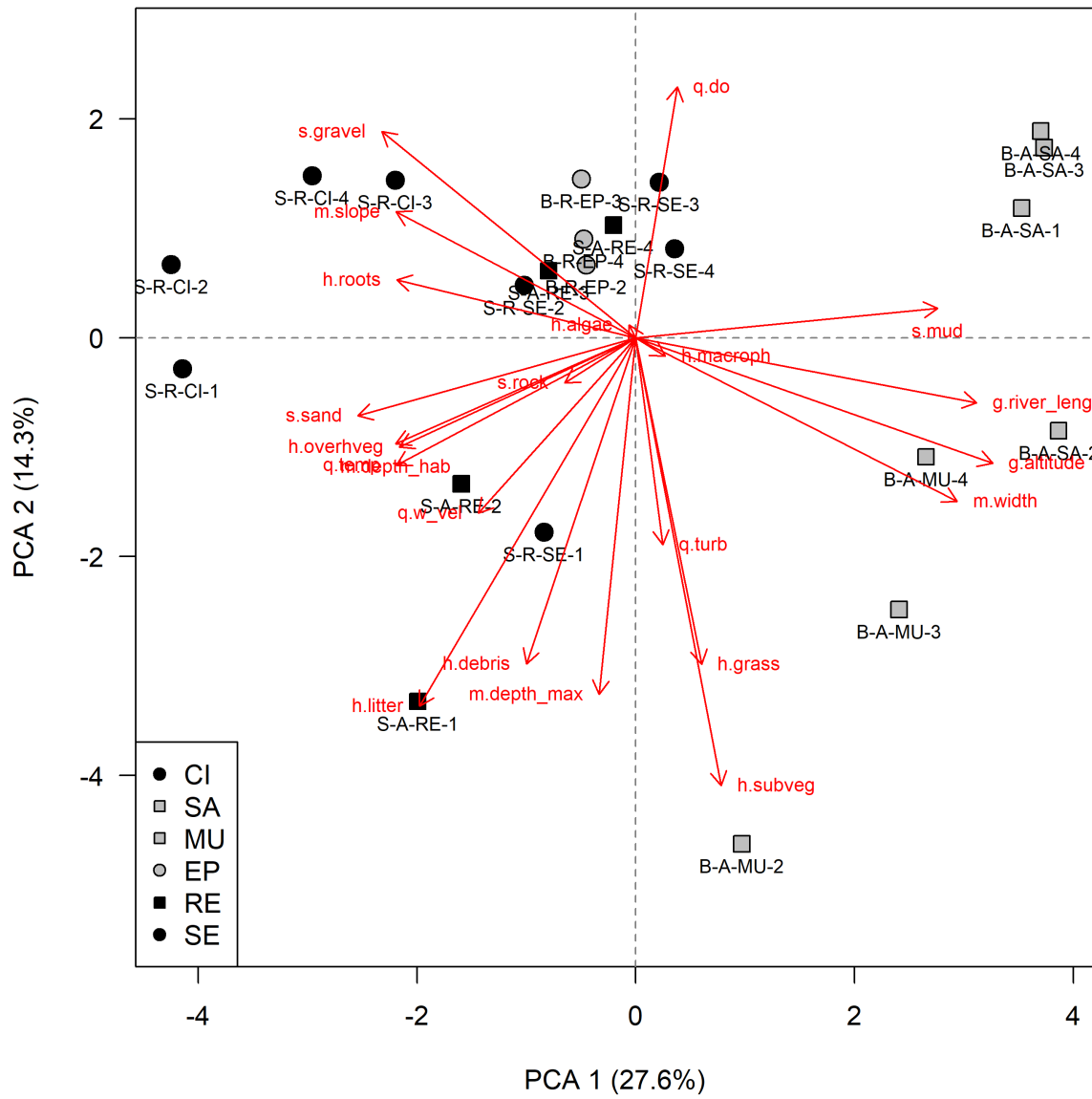


Figure 5: Environment data Principal Component Analysis using the `prcomp` and `plot` functions.

Fig.: Diversity descriptors data table

Sampling sites were different in macroinvertebrate rarefied richness (ANOVA d.f. = 5, 16; $F_{\text{richness}} = 6.9$; $p=0.001$) but not in density of individuals (ANOVA d.f. = 5, 16; $F_{\text{density}} = 2.06$; $p = 0.124$). According to *post hoc* Turkey test, Cipó stream and the Recando reservoir had significantly higher rarefied richness when compared to the other study sites ($p < 0.05$) (Figure 4).

The Stepwise Multiple Regression Model (SMRM) applied to the site morphology variables retained only stream width, with the final model explaining 15.4% of the variation (Adjusted $R^2 = 0.111$; d.f. = 1,20; $F = 3.63$, $p = 0.071$). Site width = $-0.36 (\pm 0.19)$, indicates that wider sites tend to have fewer species, although this effect was marginally significant ($p = 0.071$). River length had a strong negative effect on species richness (= -1.96 ± 0.34), explaining 63.1% of its variation (Adjusted $R^2 = 0.612$; d.f. = 1, 20; $F = 34.15$, $p < 0.001$). The SMRM that best explained the water quality variables (42.3% of the variation in rarefied species richness) included water velocity and water temperature as significant predictors (Adjusted $R^2 = 0.288$; d.f. = 4,17; $F = 3.12$, $p = 0.043$). Water velocity (= -19.97 ± 9.04) had a significant negative effect on species richness ($p = 0.041$) and water temperature (= 16.11 ± 4.97) had a significant positive effect ($p = 0.005$). Dissolved oxygen and turbidity were not significant predictors ($p > 0.05$). Among the substrate composition variables, gravel was the only significant predictor, explaining 32.8% of the variation in rarefied species richness (Adjusted $R^2 = 0.294$; d.f. = 1,20; $F = 9.76$, $p = 0.005$, = 1.85 ± 0.59). The SMRM for the habitat variables explained 58.6% of the variation in rarefied species richness (Adjusted $R^2 = 0.379$; d.f. = 7,14; $F = 2.83$, $p = 0.046$), with leaf litter (= 6.24 ± 2.33 , $p = 0.018$) and algal cover (= 0.92 ± 0.41 , $p = 0.039$) having significant positive effects on species richness. Other predictors such as roots (= 4.71 , $p = 0.058$) and littoral grass (= -1.01 , $p = 0.052$) may have been important showing marginal effects. Stepwise Multiple Regression was not performed between the environmental variables and species densities because the latter did not vary significantly across sites and sampling occasions.

Ordination analysis revealed a wide variation in community composition across sampling occasions with some degree of similarity within sites. The NMDS resulted in a non-metric fit (R^2) of 0.97, with a stress of 0.173.

Segregation in fauna composition in the studied areas (Figure 3); this result has been confirmed by MRPP, which has shown significant differences between the Seridó and Buíque regions ($A=0.14$; $p=0.0013$). However, despite not being clearly revealed in ordination, the composition has also been significantly distinct in other collection sites ($A=0.43$; $p<0.001$) and studied habitats ($A=0.07$; $p=0.04$).

The indicator species analysis revealed significant differences among all sampled sites. We also conducted a separate MRPP analysis to compare both the sampled areas (Buick and Ceridol) and the types of environments (rivers and reservoirs). This second analysis demonstrated significant differences between the areas as well as between the two types of environments. To avoid bias caused by the unequal number of samples collected in different

Descritores da diversidade por espécie (colunas). Sum, soma; RA, abundância relativa (%); mean, média; DP, desvio padrão da média; Max, maior valor; Min, menor valor; MinZ, menor valor não zero; FO, frequência de ocorrência (%) S, riqueza (ou no. de ocorrências, da matriz transposta); E, índice de equitabilidade de Pielou; H, índice de diversidade de Shannon; D, índice de diversidade de Simpson.

Spp	Sum	RA	Mean	DP	Max	Min	MinZ	FO	S	E	H	D
Thiaridae	29483.33	50.26	1340.15	3068.95	11852.08	0	2.08	50.00	11.00	0.63	1.50	0.73
Chironomidae_L	13839.58	23.59	629.07	1424.72	5997.92	0	2.08	90.91	20.00	0.52	1.56	0.73
Oligochaeta	8370.83	14.27	380.49	856.62	3045.83	0	2.08	50.00	11.00	0.60	1.44	0.73
Corixidae	1833.33	3.13	83.33	315.65	1477.08	0	43.75	18.18	4.00	0.48	0.66	0.33
Planorbidae	1493.75	2.55	67.90	148.29	489.58	0	2.08	59.09	13.00	0.63	1.62	0.75
Libellulidae	497.92	0.85	22.63	46.53	166.67	0	22.92	27.27	6.00	0.89	1.60	0.77
Hidrophilidae_L	485.42	0.83	22.06	81.26	377.08	0	2.08	27.27	6.00	0.40	0.71	0.37
Conchostraca	375.00	0.64	17.05	66.43	310.42	0	14.58	13.64	3.00	0.50	0.55	0.30
Caenidae	318.75	0.54	14.49	56.22	260.42	0	2.08	13.64	3.00	0.46	0.50	0.30
Chironomidae_P	289.58	0.49	13.16	39.22	175.00	0	4.17	22.73	5.00	0.66	1.07	0.57
Ostracoda	270.83	0.46	12.31	56.36	264.58	0	6.25	9.09	2.00	0.16	0.11	0.05
Baetidae	233.33	0.40	10.61	37.88	177.08	0	2.08	31.82	7.00	0.44	0.86	0.40
Hidrophilidae_A	204.17	0.35	9.28	23.70	106.25	0	2.08	31.82	7.00	0.73	1.42	0.67
Ceratopogonidae	156.25	0.27	7.10	18.24	83.33	0	2.08	45.45	10.00	0.66	1.53	0.67
Notonectidae	137.50	0.23	6.25	11.52	35.42	0	6.25	27.27	6.00	0.95	1.70	0.81
Atyidae	122.92	0.21	5.59	14.67	56.25	0	6.25	18.18	4.00	0.85	1.17	0.66
Hydrachnidia	93.75	0.16	4.26	19.99	93.75	0	93.75	4.55	1.00	0.00	0.00	0.00
Gomphidae	83.33	0.14	3.79	7.92	33.33	0	2.08	40.91	9.00	0.79	1.74	0.77
Leptohyphidae	72.92	0.12	3.31	15.55	72.92	0	72.92	4.55	1.00	0.00	0.00	0.00
Dytiscidae_A	70.83	0.12	3.22	8.57	37.50	0	2.08	22.73	5.00	0.78	1.26	0.65
Physidae	62.50	0.11	2.84	9.20	41.67	0	10.42	13.64	3.00	0.79	0.87	0.50
Coenagrionidae	37.50	0.06	1.70	5.81	27.08	0	2.08	18.18	4.00	0.64	0.88	0.45
Naucoridae	27.08	0.05	1.23	4.88	22.92	0	2.08	13.64	3.00	0.49	0.54	0.27
Hirudinea	16.67	0.03	0.76	2.53	10.42	0	6.25	9.09	2.00	0.95	0.66	0.47
Curculionidae	12.50	0.02	0.57	2.24	10.42	0	2.08	9.09	2.00	0.65	0.45	0.28
Glossosomatidae	12.50	0.02	0.57	1.84	6.25	0	6.25	9.09	2.00	1.00	0.69	0.50
Dytiscidae_L	12.50	0.02	0.57	1.46	6.25	0	2.08	18.18	4.00	0.90	1.24	0.67
Ampularidae	8.33	0.01	0.38	1.04	4.17	0	2.08	13.64	3.00	0.95	1.04	0.62
Leptoceridae	6.25	0.01	0.28	1.33	6.25	0	6.25	4.55	1.00	0.00	0.00	0.00
Lymnaeidae	6.25	0.01	0.28	1.33	6.25	0	6.25	4.55	1.00	0.00	0.00	0.00
Veliidae	6.25	0.01	0.28	1.33	6.25	0	6.25	4.55	1.00	0.00	0.00	0.00
Belastomatidae	4.17	0.01	0.19	0.89	4.17	0	4.17	4.55	1.00	0.00	0.00	0.00
Chaoboridae	4.17	0.01	0.19	0.89	4.17	0	4.17	4.55	1.00	0.00	0.00	0.00
Gerridae	4.17	0.01	0.19	0.89	4.17	0	4.17	4.55	1.00	0.00	0.00	0.00
Sphaeridae	2.08	0.00	0.09	0.44	2.08	0	2.08	4.55	1.00	0.00	0.00	0.00
SomaTotalD	58656.25	100.00	2666.19	6354.39	25275.00	0	360.42	745.45	164.00	17.50	27.38	14.00
MediaTotalD	1675.89	2.86	76.18	181.55	722.14	0	10.30	21.30	4.69	0.50	0.78	0.40
DPTotalD	5528.62	9.43	251.30	573.30	2238.71	0	20.01	19.31	4.25	0.35	0.61	0.30

Figure 6: Diversity descriptors of the community structure data table.

environments, we focused the analysis on comparisons between ecosystem types (e.g., rivers and reservoirs). This approach allowed us to minimize sampling imbalance while still capturing ecological differences among habitats. *Indicator Species Analysis* has shown that *Ceratopogonidae* (IV=71.2; $p=0.007$), *Chironomidae* larvae (IV=90.1; $p=0.006$), *Gomphidae* (IV=50.4; $p=0.03$), *Hydrophilidae* larvae (IV=50.0; $p=0.008$), adult *Hydrophilidae* (IV=58.3; $p=0.005$), *Libellulidae* (IV=50.0; $p=0.009$) and *Planorbidae* (IV=77.4; $p=0.004$) were indicator taxa for Seridó. Among the studied habitats, only rivers have shown significantly indicator taxa: *Atyidae* (IV=36.4; $p=0.01$), *Gomphidae* (IV=54.7; $p=0.02$) and *Thiaridae* (IV=79.8; $p=0.002$). When taking collection sites into consideration, *Ceratopogonidae* (IV=51.3; $p=0.04$), *Corixidae* (IV=68.7; $p=0.01$), *Chironomidae* larvae (IV=55.1; $p=0.02$) e *Notonectidae* (IV=42.0; $p=0.05$) were indicator taxa for Cipó river (Site 2). Whilst *Coenagrionidae* (IV=61.8; $p=0.03$), *Conchostraca* (IV=75.0; $p=0.02$), *Hydrophilidae* larvae (IV=46.3; $p=0.05$), *Libellulidae* (IV=65.5; $p=0.01$), *Oligochaeta* (IV=82.9; $p=0.001$) e *Physidae* (IV=75.0; $p=0.12$) were Recanto reservoir indicator (Site 3). Lastly, *Thiaridae* was an indicator taxa for Seridó river (Site 1) (IV=58.5; $p=0.03$) (Figure 3). No significant correlation between sediment organic matter percentage and macroinvertebrate richness ($=0.802$) or diversity ($=0.998$) has been found in the studied environments. The first three axes of the CCA have explained 38% of the spatial variance in macroinvertebrate composition (2.2 inertia). The larger portion of this variation has been explained by the first axis (16.2%). However, axes 2 and 3 have also been important in order to explain data matrix variation. Relation between benthic macroinvertebrate composition and environmental variables was significant, as shown by Monte Carlo test results for Eigenvalues ($p=0.015$) (Table II). According to intraclass correlations between environmental variables and CCA axes, the most important variables capable of explain macroinvertebrate composition were dissolved oxygen, flow, macrophytes, mud and sand (see Table II and Figure 4).

4 Discussion

5 References

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Apêndices

 Non-used codes